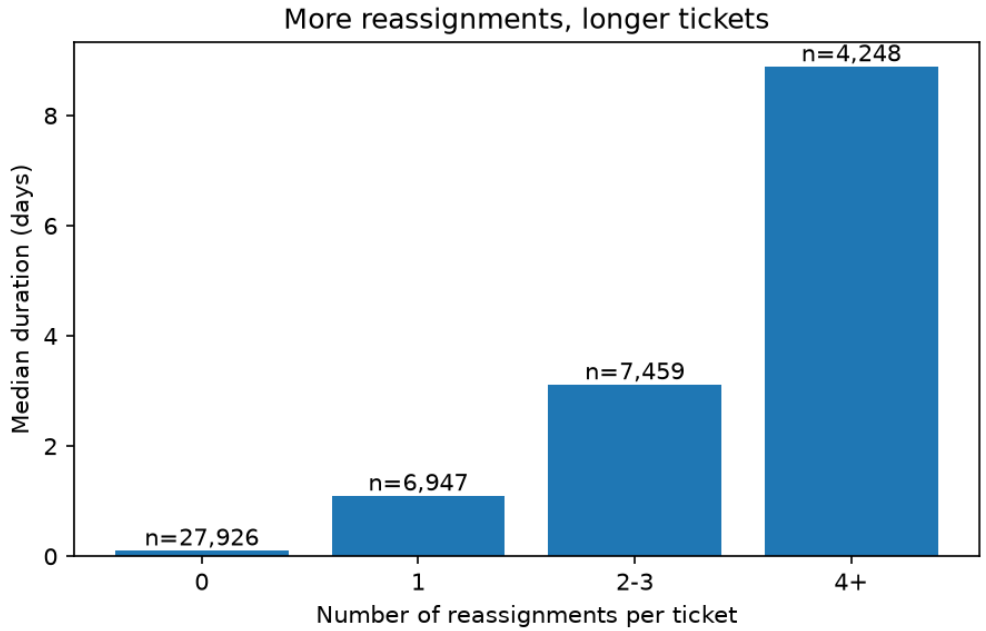
Tickets reassigned 4+ times take about 8x longer than tickets reassigned once

Process mining of 46,580 IT incident tickets (Rabobank, BPI Challenge 2014): where do tickets lose time between teams?



KEY FINDINGS

1. Rework goes with delay. 40% of tickets are reassigned at least once. Median duration: 0.1 days with none, 1.1 with one, 8.9 with 4+ (4,248 tickets, 9%).

2. The process is highly variable. Only 44.7% of tickets follow the 3 standard paths. Even a model learned from all data fits just 65.8%.

3. A few handovers stall tickets. TEAM0101 to TEAM0102 is slowest (median 94 h). TEAM0003 to TEAM0002 has the largest total impact (754 handovers, median 43 h).

RECOMMENDATION

Add a routing check at first assignment to cut reassignments, and escalate tickets waiting 24 to 48 h at slow handovers (start with the two pairs above). To be validated with the teams: the data shows where time is lost, not why.

Data: BPI Challenge 2014 (Rabobank Group ICT), 46,580 tickets, 2013-2014. Analysis: Python, pandas, PM4Py.
Caveats: association, not proven cause. Waits are calendar hours between logged core events. Standard process is self-defined.